Practice: Sampling and Variability

Module 4 | Lesson 1 | From a population to a sample

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. Identify the population, frame, observational unit, parameters, and statistics in EAI.
2. Reproduce the sample salary error and training error in percentage points.
3. Draw 50 distinct IDs from 1:2500 with a fixed seed. Explain `replace = FALSE` and why a different seed may change the result.
4. Explain why we cannot calculate a verified mean salary from these IDs alone.
5. A manager surveys only staff who volunteer. Does increasing the volunteer count guarantee removal of bias? Distinguish this issue from sampling variability.